

TOPOLOGICAL MANIFOLDS, DIFFERENTIAL
TOPOLOGY AND SPACETIME

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CHAPTER 1

TOPOLOGY AND THE THEORY OF MANIFOLDS

1.1 Topological Spaces

It is useful to recall the definition of a topology and topological space. Let X be a set (possibly empty) and $\mathcal{T}$ a collection of subsets of X . We call $\mathcal{T}$ a topology on X provided each of the following hold true

- (i) $\emptyset, X \in \mathcal{T}$.
- (ii) If $\{\mathcal{U}_\alpha : \alpha \in \mathcal{I}\}$ is an indexed collection of sets in $\mathcal{T}$ then $\bigcup_{\alpha \in \mathcal{I}} \mathcal{U}_\alpha \in \mathcal{T}$. This is called *closure under union*.
- (iii) If $\mathcal{U}, \mathcal{V} \in \mathcal{T}$ are given, then $\mathcal{U} \cap \mathcal{V} \in \mathcal{T}$. This is called *closure under finite intersections*.

Note that the name for (iii) above is no mistake. If the intersection of any two elements in $\mathcal{T}$ lies in $\mathcal{T}$, then $\bigcap_{n=1}^N \mathcal{U}_n$ for any $N \in \mathbb{N}$. To see this, one needs only argue by induction. The base case is given to us. Now assume this holds for $N \in \mathbb{N}$ and let $\mathcal{V} := \bigcap_{n=1}^N \mathcal{U}_n \in \mathcal{T}$ by hypothesis. Then, by our result for two sets:

$$\bigcap_{n=1}^{N+1} \mathcal{U}_n = \bigcap_{n=1}^N \mathcal{U}_n \cap \mathcal{U}_{N+1} = \mathcal{V} \cap \mathcal{U}_{N+1} \in \mathcal{T}$$

Given a set X and a topology on X , $\mathcal{T}$, we call the pair $(X, \mathcal{T})$ a **topological space**. The sets $\mathcal{U} \in \mathcal{T}$ are called the **open sets** of the space. The subtlety in the definition of a topology is that the collection $\mathcal{T}$ need be closed under *any* union of elements, even uncountable. This is of course motivated by the open ball topology, in particular the usual topology embedded in $\mathbb{R}$.

Before we jump into the actual mathematics let us first describe, non-rigorously, what one should expect from spacetime. This is a *mathematical* definition that underlies all of modern physics. Spacetime is what we call a 4-dimensional topological manifold, with a smooth C^∞ atlas. Of course, the experienced reader will note that we left out discussion of the Lorentzian metric, torsion and Einstein equations. As we stated above, “non-rigorously”. We shall fix this later.

The absolute lowest level definition of spacetime one can give, is to define it like $\mathbb{R}^3$. We have an *idea* that spacetime should be 4-dimensional, so one could simply define it to be $\mathbb{R}^4$, from a set-theory point of view. That is, simply the set $\{(x, y, z, t) : x, y, z \in \mathbb{R}, t \geq 0\}$. There is a fundamental flaw with this approach. We have no specific topology, and hence we cannot discuss the continuity of functions, let alone use analytic methods. Note that specifying a metric, is enough to induce a topology. It is nonetheless natural to seek out the weakest starting point, to give us enough *freedom* later on and will not harm the generality of what we seek to accomplish.

We expect the reader to have some mathematical background. If this is not the case, they can simply think of topological spaces as purely abstract objects for the moment. Otherwise, there are countless examples available online. Nonetheless, an example is in order. A metric space (X, d) induces the so called *metric topology*, where we say a set $O \subseteq X$ is open if $B(x, \delta) := \{y \in X : d(x, y) < \delta\} \subseteq O$ for some $\delta > 0$. This is the same topology one often endows $\mathbb{R}^d$ with.

Back to the notion of a topological space. A set $F \subseteq X$ is called closed if $F^c \in \mathcal{T}$ (i.e. $F^c := X \setminus F$ is an open set).

We now define the notion of a continuous mapping:

DEFINITION 1. Let $(X, \mathcal{T}_x), (Y, \mathcal{T}_y)$ be two topological spaces. A function $f : X \rightarrow Y$ is said to be *continuous* if and only if for all open sets $V \in \mathcal{T}_y$ one has $f^{-1}(V) \in \mathcal{T}_x$.

Some additional terminology needs to be introduced. A function $f : X \rightarrow Y$ is said to be **injective** if $f(x) = f(y)$ implies that $x = y$. This same function is said to be **surjective** when for all $y \in Y$ there exists $x \in X$ such that $f(x) = y$. If a map is both injective and surjective we say that it is bijective. Moreover,

DEFINITION 2. A map $f : X \rightarrow Y$ where $(X, \mathcal{T}_x), (Y, \mathcal{T}_y)$ are topological spaces, is said to be a **homeomorphism** provided it is a continuous bijection and the inverse map $f^{-1} : Y \rightarrow X$ is also continuous. In this case, we shall say that X and Y are **homeomorphic**.

In this case, we would like this definition to agree with our intuition. For instance, we would expect the “sum” of continuous maps to be again continuous. But, we are working in these abstract spaces, and we don’t always know what “addition” is! One thing we can still discuss is composition:

PROPOSITION 1.1.1. Let $(X, \mathcal{T}_x), (Y, \mathcal{T}_y), (Z, \mathcal{T}_z)$ be topological spaces and $f : X \rightarrow Y, g : Y \rightarrow Z$ continuous mappings. Then $g \circ f : X \rightarrow Z$ is continuous.

Proof. Let $O_z \subseteq Z$ be open, then $Y \supseteq g^{-1}(O_z) \in \mathcal{T}_y$ is an open set, by the continuity of g . Again, by continuity of f one sees that $f^{-1}(g^{-1}(O_z))$ is open in $(X, \mathcal{T}_x)$. But this is precisely the set $(g \circ f)^{-1}(O_z)$, which proves that $g \circ f$ is continuous.

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Now we shall briefly consider the subspace topology. Let $(X, \mathcal{T}_x)$ be a topological space and $S \subseteq X$. Is it possible to modify $\mathcal{T}_x$ to induce a topology $\mathcal{T}_s$ on S such that the pair $(S, \mathcal{T}_s)$ is a topological space? Certainly, and this is called the *subspace topology*. Define:

$$\mathcal{T}_s := \{O \cap S : O \in \mathcal{T}_x\} \quad (1.1)$$

Now, we must say the following:

PROPOSITION 1.1.2. The subspace topology together with S forms a topological space $(S, \mathcal{T}_s)$.

Proof. Clearly, $\emptyset \in \mathcal{T}_s$. To see that $S \in \mathcal{T}_s$, note that $X \in \mathcal{T}_x$ so that $S \cap X \in \mathcal{T}_s$. Now let $\{G_i\}_i$ be a collection in $\mathcal{T}_s$. For all i there is $H_i \in \mathcal{T}_x$ so that $G_i = H_i \cap S$. Then,

$$\bigcup_i G_i = \bigcup_i (H_i \cap S) = S \cap \bigcup_i H_i$$

but, $\bigcup_i H_i \in \mathcal{T}_x$ since $\mathcal{T}_x$ is a topology, and hence corresponds to a set $O \in \mathcal{T}_x$. Thus, $\bigcup_i G_i = O \cap S$ for some $O \in \mathcal{T}_x$ implying that $\bigcup_i G_i \in \mathcal{T}_s$. We only need to show that if $U, V \in \mathcal{T}_s$ then so is $U \cap V$. Again, there is $G, H \in \mathcal{T}_x$ so that $U = G \cap S$ and $V = H \cap S$. Then,

$$U \cap V = (G \cap S) \cap (H \cap S) = (G \cap H) \cap S \in \mathcal{T}_s$$

and we are done.

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A useful consequence of this is that continuity is preserved through subspaces. By this we mean, that restrictions of continuous maps to subspaces are continuous with respect to the induced subspace topology. Indeed, let $f : X \rightarrow Y$ be continuous for two topological spaces $(X, \mathcal{T}_x), (Y, \mathcal{T}_y)$. Define $\sigma : S \rightarrow Y$ to be the restriction of f to a subset $S \subseteq X$. We claim this is continuous. Let $V \in \mathcal{T}_y$ and consider $\sigma^{-1}(V)$. But, $\sigma^{-1}(V) = f^{-1}(V) \cap S$, and hence $\sigma^{-1}(V) \in \mathcal{T}_s$ by continuity of f ($f^{-1}(V) \in \mathcal{T}_x$).

Equipped with these basic, yet useful notions of topological spaces we shall move onto special topological spaces, which are common in modern physics. This is inevitable, topological spaces are fantastic worlds as they offer stupendous generality, but this is also a double edged sword. Certainly, there are so many topological spaces that it is far too likely that there will exist a topological space that will serve as an exception, no matter what we are trying to establish. To avoid this, we need to make some stronger assumptions and restrict ourselves to some better behaved spaces.

1.2 Topological Manifolds

For much of modern geometry and physics, it is sufficient to focus on these special classes of topological spaces, i.e. topological manifolds $(\mathcal{M}, \mathcal{T})$ where $\mathcal{M}$ is a manifold and $\mathcal{T}$ the topology on $\mathcal{M}$. Loosely speaking, this manifold $\mathcal{M}$ looks like Euclidean space locally. Namely, it can be covered entirely by homeomorphisms from open subsets of $\mathbb{R}^d$, for some d . More precisely, the union of the images of these homeomorphisms covers the entirety of the manifold $\mathcal{M}$. We make this rigorous below:

DEFINITION 3. A topological space $(\mathcal{M}, \mathcal{T})$ is said to be a d -dimensional **topological manifold** if for all points $p \in \mathcal{M}$ there exists an open set $\mathcal{U}$ in $\mathcal{T}_x$, containing p that is homeomorphic to some open subset of $\mathbb{R}^d$ by some homeomorphism σ .

Unfortunately, terminology is just beginning. If we have a topological manifold $(\mathcal{M}, \mathcal{T})$ then given a point p an associated pair $(\mathcal{U}, \sigma)$ is called a **chart**, as $\sigma(\mathcal{U}) \subseteq \mathbb{R}^d$. Then, an indexed collection of charts $\mathcal{A} := \{(\mathcal{U}_\alpha, \sigma_\alpha) : \alpha \in I\}$, for some index set I (not necessarily countable), is called an **atlas** of $(\mathcal{M}, \mathcal{T})$ if and only if $\mathcal{M} = \bigcup_{\alpha \in I} \mathcal{U}_\alpha$. Such a σ (homeomorphism) is called a **chart-mapping** or chart-map. Now, as σ maps into $\mathbb{R}^d$ we may also write $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_d)$ where each $\sigma_j : \mathcal{U} \rightarrow \mathbb{R}$. These σ_j are called the **coordinate maps** of the chart. $\sigma_j(p)$ is called the j^{th} **coordinate of p** .

Finally, we must study transition maps. Let $p \in \mathcal{M}$ and consider two charts $(\mathcal{U}, \sigma)$ and $(\mathcal{V}, \tau)$ on $\mathcal{M}$. Recall that σ, τ are homeomorphisms and suppose that $\mathcal{U} \cap \mathcal{V} \neq \emptyset$ (in which case they are open). Then, $\sigma : \mathcal{U} \rightarrow \mathbb{R}^d$ and $\tau : \mathcal{V} \rightarrow \mathbb{R}^d$. Consider now the restriction: $\tau \circ \sigma^{-1} : \sigma(\mathcal{U}) \rightarrow \tau(\mathcal{V})$. By our previous propositions we see that this mapping is again a homeomorphism, and takes a point in the image of $\mathcal{U} \cap \mathcal{V}$ in $\sigma(\mathcal{U})$ to its corresponding point in $\tau(\mathcal{V})$. This is called the **transition map** from x to y .

Now, a curve γ on $\mathcal{M}$ is a mapping from some subset (an interval) of the real line $\gamma : \mathbb{R} \rightarrow \mathcal{M}$. We shall study a portion of the curve, primarily. Let $p \in \mathcal{M}$ be a point on the curve, and let $(\mathcal{U}, \sigma)$ be a chart for this point, on $\mathcal{M}$. We say this curve is *continuous* if $\sigma \circ \gamma : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^d$ is itself continuous. Sounds good? No, it is not. There is a subtlety here, we must show that this concept is *well-defined*. That is, we must show that it is independent of the chosen chart. It would not make sense to call this curve continuous if one could pick up a chart for which γ were continuous, but another that would contradict this. As such, we rid ourselves of this issue with the following lemma:

LEMMA 1.2.1. Let $(\mathcal{M}, \mathcal{T})$ be a topological manifold and $\gamma : I \subseteq \mathbb{R} \rightarrow \mathcal{M}$ a curve in $\mathcal{M}$. If $(\mathcal{U}, \sigma)$ and $(\mathcal{V}, \tau)$ are charts and γ is continuous with respect to $(\mathcal{U}, \sigma)$, it is also continuous with respect to $(\mathcal{V}, \tau)$. That is, continuity is independent of chart.

Proof. Since γ is continuous with respect to the chart $(\mathcal{U}, \sigma)$ we find that the map $\sigma \circ \gamma : I \subseteq \mathbb{R} \rightarrow \mathbb{R}^d$ is a continuous mapping. Now, consider $\tau \circ \gamma : I \rightarrow \mathbb{R}^d$. We may write:

$$\tau \circ \gamma = \tau \circ (\sigma^{-1} \circ \sigma) \circ \gamma = (\tau \circ \sigma^{-1}) \circ (\sigma \circ \gamma)$$

which we know to be continuous, as the composition of continuous maps.

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This is a small step, ultimately. We wish to do the same for a stronger class of manifolds, differential and smooth manifolds. In light of what we have done above, it would make sense to try and define a curve on a manifold to be smooth if $\sigma \circ \gamma$ were differentiable (or smooth) for a chart $(\mathcal{U}, \sigma)$. Unfortunately, this would not be well defined, and this is because requiring differentiability is much stronger. The “trick” used above won’t work either, if $\sigma : \mathcal{U} \rightarrow \mathbb{R}^d$ is differentiable there is no guarantee that $\sigma^{-1} : \sigma(\mathcal{U}) \rightarrow \mathcal{U}$ would also be.

Hence, we have some work ahead of us. Before we tackle the problem of differentiable and smooth manifolds, we shall briefly cover some material in Tensor and Multilinear algebra that will serve us when we discuss differential forms and integration on manifolds. This is the next section.

The reader may chose to skip the subsequent section if they are comfortable or simply with to take a simplistic approach.

1.3 Multilinear Algebra and Exterior Algebras

The entity of discussion in this section are essentially vector spaces. Recall that a vector space over $\mathbb{C}$, the field is complex numbers, is a set $\mathcal{V}$ of vectors equipped with two operations:

$$\oplus : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}, \quad (u, v) \mapsto u \oplus v \quad (1.2)$$

$$\odot : \mathbb{C} \times \mathcal{V} \rightarrow \mathcal{V}, \quad (\zeta, u) \mapsto \zeta \odot u \quad (1.3)$$

Such that $\mathcal{V}$ is an Abelian group with respect to both structures. We shall often replace the exotic $\oplus, \odot$ notation by $+, \cdot$ when the context is clear. Although in this text the main goal is to develop abstract mathematical structures that are useful in the study of differential and analytic topology (such as smooth manifolds), a vector space is an *auxiliary structure* in this context. The notion of a vector space will allow us to discuss *multilinear mappings*, construct Tangent and Cotangent spaces, but we not develop spacetime as a vector space.

Certainly, it does not make sense to endow space-time with a structure as above. What does it mean to “add” two points in spacetime? To multiply a point in time by a scalar? We shall view it purely as an abstract manifold (with a little more stuff, which we defer until later on). We first study vector spaces *abstractly* to develop a good understanding of *tensors*.

From here on we write $\mathcal{V}, \mathcal{W}$ to denote arbitrary vector spaces over $\mathbb{C}$. A mapping $T : \mathcal{V} \rightarrow \mathcal{W}$ is called a **linear** mapping provided that $T(u + v) = T(u) + T(v)$ for all $u, v \in \mathcal{V}$ and $T(\zeta u) = \zeta T(u)$ for all $u \in \mathcal{V}$ and $\zeta \in \mathbb{C}$. We cannot discuss whether these mappings are continuous, as we have not a specified topology. Similarly, we cannot discuss linearity in a topological space without endowing the space with the two operations given above.

To ease notation, we shall write $T : \mathcal{V} \xrightarrow{\sim} \mathcal{W}$ to say that T is a linear mapping from $\mathcal{V}$ to $\mathcal{W}$. Moreover, for the sake of abstraction we shall consider first vector spaces over an arbitrary field $\mathbb{F}$.

1.3–1.3.1 Spanning Sets and Bases

Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$ and $\mathbb{S} \subseteq \mathcal{V}$ a non-empty subset. We may describe $\mathbb{S}$ as the collection: $\mathbb{S} = \{v_i \in \mathcal{V} : i \in I\}$ where I is some index set (not necessarily countable). A **linear combination** of elements in $\mathbb{S}$ is a combination of the form:

$$\alpha_1 v_{i_1} + \alpha_2 v_{i_2} + \cdots + \alpha_n v_{i_n}, \quad i_j \in I \text{ and } \alpha_j \in \mathbb{F} \quad \forall j \quad (1.4)$$

Of course, in the above $+$ describes the vector addition that the space $\mathcal{V}$ is equipped with and $\alpha_j v_j$ utilizes the embedded scalar multiplication over $\mathbb{F}$. We must *stress* that the combination above contain **only finitely many** combinations (i.e. there are only finitely many elements being “summed”). The reason for this is to guarantee that any such linear combination results in an element $u \in \mathcal{V}$, as we only have closure of $\mathcal{V}$ under finitely many such operations. We have the following elementary definition:

DEFINITION 4. Let $W \subseteq \mathcal{V}$ be a non-empty set. We call W a **vector subspace** if and only if:

(i) $u + v \in W$ whenever $u, v \in W$.

(ii) $\alpha v \in W$ whenever $v \in W$ and $\alpha \in \mathbb{F}$.

To emphasize that W is a subspace, we shall write $\mathcal{W}$ and/or $W \subseteq \mathcal{V}$.

If $\mathbb{S} \subseteq \mathcal{V}$ is a given set, non-empty, we may define the **spanning set** of $\mathbb{S}$, which we shall denote by $\text{span}(\mathbb{S})$. This consists of the collection of all linear combinations of elements in $\mathbb{S}$ over the field $\mathbb{F}$. In set builder notation:

$$\text{span}(\mathbb{S}) := \left\{ \sum_{j=1}^n \alpha_j u_j : \alpha_j \in \mathbb{F}, u_j \in \mathbb{S} \right\} \quad (1.5)$$

and $n \in \mathbb{N}$ may vary to exhaust $\mathbb{S}$. Note that even in the case where $\mathbb{S}$ is empty, $0 \in \text{span}(\mathbb{S})$ by definition of the empty sum. When $\mathbb{S} \neq \emptyset$ by taking $\alpha_j = 0$ for all j up to some n we have $0 \in \text{span}(\mathbb{S})$ anyways.

If $\mathbb{S}$ is non-empty and a subset of $\mathcal{V}$, then $\text{span}(\mathbb{S})$ is a subspace of $\mathcal{V}$. Indeed, if $u, v \in \text{span}(\mathbb{S})$ then we may write

$$u = \sum_{j=1}^n \alpha_j u_j, \quad v = \sum_{j=1}^n \beta_j u_j$$

This may always be done, by setting $\alpha_j, \beta_j = 0$ whenever some u_j does not appear in the linear combination. Having done this, we may write now

$$u + v = \sum_{j=1}^n (\alpha_j + \beta_j) u_j = \sum_{j=1}^n \gamma_j u_j \in \text{span}(\mathbb{S}), \quad \gamma_j := \alpha_j + \beta_j \in \mathbb{F}$$

Finally, let $u \in \text{span}(\mathbb{S})$ and $\delta \in \mathbb{F}$ be given. Then, since $u = \sum_{j=1}^n \alpha_j u_j$ we find that $\delta u = \sum_{j=1}^n \delta \alpha_j u_j = \sum_{j=1}^n \varepsilon_j u_j$ where $\varepsilon_j \in \mathbb{F}$ by the same argument, whence $\delta u \in \text{span}(\mathbb{S})$ and thus $\text{span}(\mathbb{S})$ is a subspace of $\mathcal{V}$.

DEFINITION 5. Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ and $\mathbb{S} \subseteq \mathcal{V}$ non-empty. We say that $\mathbb{S}$ is a **linearly dependent family** if there exists a sub-collection $\{u_1, u_2, \dots, u_n\} \subseteq \mathbb{S}$ and coefficients $\{\alpha_1, \alpha_2, \dots, \alpha_n\}$, at least one $\alpha_j \neq 0$, such that

$$\sum_{j=1}^n \alpha_j u_j = 0$$

If no such family exists, then we call $\mathbb{S}$ **linearly independent**.

An example of a dependent family is any $\mathbb{S} \subseteq \mathcal{V}$ containing the 0 vector. An example of an independent family is the collection $\{(1, 0), (0, 1)\}$ in $\mathbb{R}^2$. It now time for our first formal result, which is quite easy.

PROPOSITION 1.3.1. Let $\mathcal{V}$ be a vector space and $\mathbb{S} \subseteq \mathcal{V}$ non-empty. $u \in \text{span}(\mathbb{S})$ if and only if $\text{span}(\mathbb{S}) = \text{span}(\mathbb{S} \cup \{u\})$.

Proof. ($\implies$) Let $u \in \text{span}(\mathbb{S})$. We clearly have $\text{span}(\mathbb{S}) \subseteq \text{span}(\mathbb{S} \cup \{u\})$. Let now $v \in \text{span}(\mathbb{S} \cup \{u\})$. We may write v as the linear combination of elements in $\mathbb{S}$ and possibly u . But, u is in $\mathbb{S}$ and hence this is also a linear combination of elements in $\mathbb{S}$, i.e. $v \in \text{span}(\mathbb{S})$ which establishes the desired equality.

($\impliedby$) Suppose $\text{span}(\mathbb{S}) = \text{span}(\mathbb{S} \cup \{u\})$, we show $u \in \text{span}(\mathbb{S})$. This is clear from the observation that $u \in \text{span}(\mathbb{S} \cup \{u\})$ (why?) and, by equality, $u \in \text{span}(\mathbb{S})$.

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DEFINITION 6. A collection of vectors $\mathbb{S}$ in $\mathcal{V}$ is called **maximally linearly independent** (or **maximally independent**) if $\mathbb{S}$ is independent and for any $v \in \mathcal{V}$ the union $\mathbb{S} \cup \{v\}$ is dependent. We say that $\mathbb{S}$ **spans** $\mathcal{V}$ if and only if $\text{span}(\mathbb{S}) = \mathcal{V}$. In this case, we write that $\mathbb{S}$ is **minimally spanning** if $T \subsetneq \mathbb{S}$ implies $\text{span}(T) \subsetneq \mathcal{V}$.

The following theorem is a beautiful, and crucial characterization of such spanning sets:

THEOREM 1.3.2. Let $\mathcal{V}$ be a vector space and $\mathbb{S} \subseteq \mathcal{V}$ non-empty. The following statements are equivalent:

- (i) $\mathbb{S}$ is a minimal spanning set.
- (ii) $\mathbb{S}$ is maximally independent.
- (iii) Each vector $u \in \mathcal{V}$ may be expressed as a unique linearly combinations of vectors in $\mathbb{S}$.

Proof. (i $\implies$ ii) Suppose that $\mathbb{S}$ is minimally spanning $\mathcal{V}$. We first show that $\mathbb{S}$ is an independent family. Suppose for some combination of elements $\{u_1, \dots, u_k\}$ in $\mathbb{S}$ we have

$$\sum_{j=1}^k \alpha_j u_j = 0, \quad \alpha_m \neq 0$$

For such an index m , we multiply through by α_m^{-1} to see that u_m is a linear combination of the other elements in the set $\mathbb{S}$, contradicting the fact that $\mathbb{S}$ is minimal spanning since $u_m \in \text{span}(\mathbb{S} \setminus \{u_m\})$ which implies by Proposition 1.3.1 that $\text{span}(\mathbb{S} \setminus \{u_m\}) = \text{span}(\mathbb{S})$. In other words, $\mathbb{S}$ is not minimal spanning, a contradiction.

Now, we show that $\mathbb{S}$ is maximal. Let $v \in \mathcal{V}$ be such that $v \notin \mathbb{S}$, we shall show that $\mathbb{S} \cup \{v\}$ is dependent. Now, $v \in \mathcal{V} = \text{span}(\mathbb{S})$ implies directly that

$$v = \alpha_1 u_1 + \cdots + \alpha_n u_n$$

If v were the 0 vector, adding it to $\mathbb{S}$ would immediately kill independence, hence we may suppose without harm $v \neq 0$; so that some $\alpha_j \neq 0$. Then write

$$0 = \alpha_1 u_1 + \cdots + \alpha_n u_n - v$$

where some coefficients are non-zero. Thus, $\mathbb{S}$ is maximally independent.

(ii $\implies$ i) Assume $\mathbb{S}$ is maximally independent. Clearly, the 0 vector has a unique linear combination of elements in $\mathbb{S}$ (take all coefficients to be 0). Consider now $v \in \mathcal{V} \setminus \{0\}$. Suppose by way of contradiction that v is not a linear combination of elements in $\mathbb{S}$. We claim $\mathbb{S} \cup \{v\}$ would be linearly independent. Indeed, if this family were instead dependent we could write

$$\alpha_1 u_1 + \cdots + \alpha_n u_n + \beta v = 0$$

where at least one coefficient must be non-zero. We observe that $\beta \neq 0$, for otherwise be the independence of $\mathbb{S}$ we would have $\alpha_1, \alpha_2, \dots, \alpha_n = 0$. Then, write

$$v = -\beta^{-1} (\alpha_1 u_1 + \cdots + \alpha_n u_n)$$

which is impossible since we assumed v was not a linear combination of elements in $\mathbb{S}$. Hence, $\mathbb{S} \cup \{v\}$ must be linearly independent. But this contradicts the assumption that $\mathbb{S}$ is maximal. We now have existence of representation. We now show uniqueness. Suppose $v \in \mathcal{V}$ has two expansions: $\sum_{j=1}^n \alpha_j u_j = \sum_{j=1}^n \beta_j u_j$, where we take $\alpha_j = 0$ or $\beta_j = 0$ as necessary to include the same vectors. Then,

$$\sum_{j=1}^n (\beta_j - \alpha_j) u_j = 0 \implies \beta_j - \alpha_j = 0$$

by independence. Restriction to $\beta_j \neq 0$ yields the desired uniqueness of representation.

(iii $\implies$ i) We need to show that $\mathbb{S}$ is minimally spanning. The fact that $\text{span}(\mathbb{S}) = \mathcal{V}$ is given to us. The uniqueness is what gives us the minimality of $\mathbb{S}$. Certainly, let $T \subsetneq \mathbb{S}$ and take an element $v \in \mathbb{S}$ but not in T . It is clear that $1 \cdot v$ is a linear combination of elements in $\mathbb{S}$ equal to v . However, by uniqueness this can be the only representation of this v in $\mathbb{S}$. Since $T \subsetneq \mathbb{S}$ there can be no representation of v in $\text{span}(T) \subseteq \text{span}(\mathbb{S})$. Hence, $\mathbb{S}$ is minimally spanning for $\mathcal{V}$.

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Wow, that was long! Onto the characterization of the spaces *themselves*!

In this section we explore the concept of a basis for a vector space. These are *extremely* important in linear algebra, as a basis for a space induces a complete characterization of the space. In fact, with the help of this new concept we are able to define the concept of *dimension* unambiguously, and as such we shall study simple dimensional theory here.

We must, of course, first define a basis

DEFINITION 7. Let $\mathcal{V}$ be a vector space over $\mathbb{F}$, and $\mathcal{B} \subseteq \mathcal{V}$ an indexed collection of vectors in $\mathcal{V}$. We call $\mathcal{B}$ a **basis** for the vector space $\mathcal{V}$ if it is minimally spanning.

Observe that we could have equivalently defined $\mathcal{B}$ to be a maximally independent subset of $\mathcal{V}$, or even a set for which any element in $\mathcal{V}$ had a unique linear representation in $\mathbb{S}$, by Theorem 1.3.2. We actually say the following:

THEOREM 1.3.3. $\mathcal{B}$ is a basis for $\mathcal{V}$ if and only if $\mathcal{B}$ is linearly independent and spanning.

Proof. ($\implies$) Let $\mathcal{B}$ be a basis for $\mathcal{V}$. Since $\mathcal{B}$ is by definition minimally spanning, we see $\text{span}(\mathcal{B}) = \mathcal{V}$. To see the independence (linearly) of $\mathcal{B}$ we apply Theorem 1.3.2 to see that $\mathcal{B}$ is also, in fact, maximally independent.

($\impliedby$) Let $\mathcal{B} \subseteq \mathcal{V}$ be linearly independent and spanning. We show that $\mathcal{B}$ is minimally spanning. Pick $u \in \mathcal{B}$ (then $u \neq 0$) and define $\mathbb{S} := \mathcal{B} \setminus \{u\}$. Note that $u \notin \text{span}(\mathbb{S})$, for otherwise

$$0 \neq u = \alpha_1 u_1 + \dots + \alpha_n u_n$$

implies that some coefficient is non-zero, and thus $0 = \alpha_1 u_2 + \dots + \alpha_n u_n - u$ contradicts the independence of $\mathcal{B}$. Hence, $\mathcal{B}$ is minimally spanning and is a basis. ○

We shall now require a very powerful, yet technical, lemma due to Steinitz:

LEMMA 1.3.4. *Let $\mathbb{S} := \{u_1, u_2, \dots\}$ be a family of vectors in $\mathcal{V}$. $\mathbb{S}$ is linearly dependent if and only if some vector in $\mathbb{S}$ is the linear combination of those preceding it.*

Proof. ($\implies$) If $\mathbb{S}$ is linearly dependent then for some sub-collection we have, after adding 0 coefficients,

$$\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_n u_n = 0, \quad \alpha_j \neq 0$$

for at least one index j . Let m denote the largest such index in the above list, then all subsequent indices are 0 so that

$$\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_m u_m = 0, \quad \alpha_m \neq 0$$

Whereby it is easily seen that

$$u_m = -\alpha_m^{-1} (\alpha_1 u_1 + \alpha_2 u_2 + \dots + \alpha_{m-1} u_{m-1})$$

Conversely, if u_n is some linear combination of the vectors preceding it the family is linearly dependent simply by writing

$$u_n = \sum_{j=1}^{n-1} \alpha_j u_j \implies 0 = \sum_{j=1}^{n-1} \alpha_j u_j - u_n$$

○

We now give this strong lemma:

LEMMA 1.3.5 (Steinitz). *Let $\mathcal{V}$ be a vector space and $\{v_1, v_2, \dots, v_n\}, \{w_1, w_2, \dots, w_m\}$ with $n \leq m$ be two independent subsets of $\mathcal{V}$. For all $0 \leq j \leq n$ we may relabel the w 's so that:*

$$\{v_1, v_2, \dots, v_j, w_{j+1}, w_{j+2}, \dots, w_m\}$$

is independent (linearly).

Proof. If $j = 0$ this set is $\{w_1, \dots, w_m\}$ and there is nothing to do. By way of induction, assume this for $0 \leq j \leq n$. If it holds for $j = n$ we are done, so without loss of generality let $j < n$. That is, we may relabel the w 's so that

$$\{v_1, v_2, \dots, v_j, w_{j+1}, w_{j+2}, \dots, w_m\}$$

is independent. Now, we adjoin v_{n+1} :

$$\{v_1, v_2, \dots, v_j, v_{j+1}, w_{j+1}, w_{j+2}, \dots, w_m\}$$

if this is linearly independent, pull out w_{n+1} and we are done. Else, some vector is a linear combination of those preceding it. After a moments reflection, it cannot be some v_j for $\{v_j\}_{j=1}^n$ is linearly independent. That is, there exists some $r \geq 1$ such that

$$w_{j+r} = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_{j+1} v_{j+1} + \beta_1 w_{j+1} + \dots + \beta_{j+r-1} w_{j+r-1} \quad (\dagger)$$

Now consider $\{v_1, v_2, \dots, v_j, v_{j+1}, w_{j+1}, w_{j+2}, \dots, \widehat{w_{j+r}}, \dots, w_m\}$. We claim that this is an independent family. Else,

$$0 = \gamma_1 v_1 + \gamma_2 v_2 + \dots + \gamma_j v_j + \gamma_{j+1} v_{j+1} + \delta_1 w_{j+1} + \dots + \delta_{j+r} \widehat{w_{j+r}} + \dots + \delta_m w_m$$

Since the collection with v_{j+1} omitted is a subset of our induction hypothesis set, we see $\gamma_{j+1} \neq 0$ and hence we may isolate v_{j+1} in terms of the others above. Substitution in (†) shows that w_{j+r} is a linear combination of the other vectors (we have expressed v_{j+1} in terms of the other vectors). Hence, our induction hypothesis was false. Contradiction.

○

We are now prepared to define the dimension of a vector space. Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$ and $\mathcal{B}$ a basis for $\mathcal{V}$. We say $\mathcal{V}$ has dimension $|\mathcal{B}|$, the cardinality of $\mathcal{B}$, and write $\dim(\mathcal{V}) = |\mathcal{B}|$. An important fact is that this is well defined.

THEOREM 1.3.6. *Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ with $\dim(\mathcal{V}) = n < \infty$. Any other basis has n elements.*

Proof. Let $\mathcal{B}$ be a basis for $\mathcal{V}$ with $\mathcal{B} := \{b_1, b_2, \dots, b_n\}$ and $\mathfrak{B} := \{c_1, c_2, \dots, c_m\}$ another basis. We shall first show that $m > n$ is impossible. If $m > n$ then by Steinitz's lemma we may take $j = n$ and relabel the elements of $\mathfrak{B}$ such that the set $\{b_1, b_2, \dots, b_n, c_{n+1}, \dots, c_m\}$ is linearly independent: contradicting the fact that $\mathcal{B}$ is maximally independent. A symmetric argument shows that $m < n$ is impossible, whence $m = n$.

○

Similarly,

COROLLARY 1.3.7. *If $\mathcal{V}$ has $\dim(\mathcal{V}) < \infty$ and $T = \{t_1, t_2, \dots, t_m\} \subseteq \mathcal{V}$ is an independent family, then $m \leq \dim(\mathcal{V})$.*

Proof. Else, we repeat the argument in the previous proof, giving that a basis $\mathcal{B}$ is not maximally independent. This same argument shows that any independent subset of a vector space $\mathcal{V}$ with $\dim(\mathcal{V}) < \infty$ can be completed to a basis.

○

The following is useful:

PROPOSITION 1.3.8. *Let $\mathcal{W} \subseteq \mathcal{V}$ be a subspace with $\dim(\mathcal{V}) < \infty$. Then, $\dim(\mathcal{W}) \leq \dim(\mathcal{V})$ and*

$$\mathcal{W} = \mathcal{V} \iff \dim(\mathcal{W}) = \dim(\mathcal{V})$$

Proof. Let $\mathcal{W}$ be a basis for $\mathcal{W}$. It is linearly independent in $\mathcal{V}$ (as a subset of $\mathcal{V}$) and hence we see by a previous lemma that $|\mathcal{W}| \leq \dim(\mathcal{V})$ which shows that $\dim(\mathcal{W}) \leq \dim(\mathcal{V})$.

Now suppose $\dim(\mathcal{V}) = \dim(\mathcal{W})$ and assume for a contradiction that $\mathcal{W} \neq \mathcal{V}$, i.e. $\mathcal{W} \subsetneq \mathcal{V}$. There exists $v \in \mathcal{V}$ such that $v \notin \mathcal{W}$. In particular, if $\mathcal{B} = \{u_1, \dots, u_n\}$ is a basis for $\mathcal{W}$, it is independent, and so is $\{u_1, u_2, \dots, u_n, v\}$ since v is not a linear combination of the vectors preceding it. This contradicts the fact that any independent set in $\mathcal{V}$ must have less than n elements.

○

Change of Basis Since a basis of a vector space is in general not uniquely determined, and it makes sense to discuss the concept of a *change of basis*. Consider a vector space $\mathcal{V}$ over a field $\mathbb{F}$. Let $\mathcal{B} = \{b_1, b_2, \dots, b_n\}, \mathcal{C} = \{c_1, c_2, \dots, c_n\}$ be two bases for $\mathcal{V}$. Suppose that $u \in \mathcal{V}$ may be represented as

$$u = \sum_{j=1}^n \alpha_j b_j$$

then we say u has **coordinates** $\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}$ with respect to the basis $\mathcal{B}$. We then denote this column-vector by $[u]_{\mathcal{B}}$. This quantity will almost always be different with respect to another basis $\mathcal{C}$, say, $[u]_{\mathcal{B}} \neq [u]_{\mathcal{C}}$ in general. A simple example is the standard basis $S_2 := \{(1, 0), (0, 1)\}$ in $\mathbb{R}^2$. The vector (π, e) would have coordinates $[(\pi, e)]_{S_2} = \begin{pmatrix} \pi \\ e \end{pmatrix}$.

The following question arises: *how do we go from $[u]_{\mathcal{B}}$ to $[u]_{\mathcal{C}}$ in general?* The answer lies within matrices. We already know that the individual basis elements of $\mathcal{C}$, the c_j 's may be written as a linear combination of elements in $\mathcal{B}$:

$$c_j = m_{1,j}b_1 + m_{2,j}b_2 + \dots + m_{n,j}b_n$$

Or, as a system:

$$\begin{aligned} c_1 &= m_{1,1}b_1 + m_{2,1}b_2 + \dots + m_{n,1}b_n \\ c_2 &= m_{1,2}b_1 + m_{2,2}b_2 + \dots + m_{n,2}b_n \\ &\vdots \\ c_n &= m_{1,n}b_1 + m_{2,n}b_2 + \dots + m_{n,n}b_n \end{aligned}$$

We now define the *transformation matrix* taking $\mathcal{B}$ to $\mathcal{C}$:

$$M_{\mathcal{B}}^{\mathcal{C}} := \begin{pmatrix} m_{1,1} & m_{1,2} & \dots & m_{1,n} \\ m_{2,1} & \dots & m_{2,n-1} & m_{2,n} \\ \vdots & \ddots & \dots & \vdots \\ m_{n,1} & m_{n,2} & \dots & m_{n,n} \end{pmatrix} \quad (1.6)$$

The following properties are immediate:

- $[v + w]_{\mathcal{B}} = [v]_{\mathcal{B}} + [w]_{\mathcal{B}}$
- $[\alpha v]_{\mathcal{B}} = \alpha[v]_{\mathcal{B}}$.

Now, we wish to prove the following:

THEOREM 1.3.9. *If $u \in \mathcal{V}$ and $\mathcal{B}, \mathcal{C}$ are as above:*

$$[u]_{\mathcal{C}} = M_{\mathcal{B}}^{\mathcal{C}}[u]_{\mathcal{B}}$$

Proof. By the linearity of $[\cdot]_{\mathcal{B}}$ we need only show this for a basis element $b_j \in \mathcal{B}$. We consider this as follows:

$$M_{\mathcal{B}}^{\mathcal{C}}[b_j]_{\mathcal{B}} = M_{\mathcal{B}}^{\mathcal{C}} \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = j^{th} \text{ column of } M_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} m_{1,j} \\ m_{2,j} \\ \vdots \\ m_{j,j} \\ \vdots \\ m_{n,j} \end{pmatrix} = [b_j]_{\mathcal{C}}$$

○

LEMMA 1.3.10. *Let M be a matrix such that $[u]_{\mathcal{C}} = M[v]_{\mathcal{B}}$ for every $u \in \mathcal{V}$. Then $M = M_{\mathcal{B}}^{\mathcal{C}}$. That is, this transformation is uniquely determined.*

Proof. Let $u \in \mathcal{V}$. By our previous theorem we know that $[u]_{\mathcal{C}} = M_{\mathcal{B}}^{\mathcal{C}}[u]_{\mathcal{B}}$. In particular, for a basis element b_j :

$$\begin{pmatrix} m_{1,j} \\ m_{2,j} \\ \vdots \\ m_{j,j} \\ \vdots \\ m_{n,j} \end{pmatrix} = M_{\mathcal{B}}^{\mathcal{C}}[u]_{\mathcal{B}} = [u]_{\mathcal{C}} = M[u]_{\mathcal{B}} = j^{th} \text{ column of } M$$

○

COROLLARY 1.3.11. *If $\mathcal{B}, \mathcal{C}, \mathcal{D}$ are bases for $\mathcal{V}$:*

1. $M_{\mathcal{B}}^{\mathcal{C}} = I \in \mathbb{F}^{n \times n}$
2. $M_{\mathcal{B}}^{\mathcal{D}} = M_{\mathcal{C}}^{\mathcal{D}} M_{\mathcal{B}}^{\mathcal{C}}$.
3. $M_{\mathcal{C}}^{\mathcal{B}} = (M_{\mathcal{B}}^{\mathcal{C}})^{-1}$.

Proof. We first show (1). This is really a consequence of the previous lemma. It is clear that I is a matrix such that $[u]_{\mathcal{B}} = I[u]_{\mathcal{B}}$ for all $u \in \mathcal{V}$, since this matrix is uniquely determined we deduce that $I = M_{\mathcal{B}}^{\mathcal{B}}$.

For (2) the argument is similar. If $u \in \mathcal{V}$ then $[u]_{\mathcal{D}} = M_{\mathcal{C}}^{\mathcal{D}}[u]_{\mathcal{C}} = M_{\mathcal{C}}^{\mathcal{D}}(M_{\mathcal{B}}^{\mathcal{C}}[u]_{\mathcal{B}})$, whence the result follows by the previous lemma.

Finally, we are left with (3). This uses successive applications of (2). Note that $[u]_{\mathcal{B}} = M_{\mathcal{C}}^{\mathcal{B}} M_{\mathcal{B}}^{\mathcal{C}}[u]_{\mathcal{B}}$ which implies by (1) that $M_{\mathcal{C}}^{\mathcal{B}} M_{\mathcal{B}}^{\mathcal{C}} = I$ and hence that they are each other's inverses.

○

1.3–1.3.2 Linear Transformations Between Spaces

We adopt again the notation of the previous sections. We begin with a crucial definition:

DEFINITION 8. *Let $\mathcal{V}, \mathcal{W}$ be vector spaces over $\mathbb{F}$ and $T : \mathcal{V} \rightarrow \mathcal{W}$ a mapping. We call T a **linear transformation** if and only if*

- (a) $T(u + v) = T(u) + T(v)$ for all $(u, v) \in \mathcal{V} \times \mathcal{V}$.
- (b) $T(\alpha u) = \alpha T(u)$ for all $\alpha \in \mathbb{F}$ and $u \in \mathcal{V}$.

Some observations are in order. Note that this simply a generalization of the smooth linear functions $f : \mathbb{R} \rightarrow \mathbb{R}$ of the form $f(x) = ax + b$ you've all seen in kindergarten. More importantly, since $0 + 0 = 0$ we take note that $T(0) = T(0 + 0) = T(0) + T(0)$ which implies $0 = T(0)$, i.e. $0 \xrightarrow{T} 0$. Similarly, $T(-u) = -T(u)$ by “pulling out” $\alpha = -1$. We define further the **kernel** of a linear transformation, denoted $\ker T$, to be the set:

$$\ker T := \{u \in \mathcal{V} : T(u) = 0 \in \mathcal{W}\} \quad (1.7)$$

An important fact is that $\ker f$ is a subspace of $\mathcal{V}$. Indeed, if $u, v \in \ker T$ then $T(u) = T(v) = 0$. Moreover, $T(u+v) = T(u) + T(v) = 0 + 0 = 0$. Let $\alpha \in \mathbb{F}$ and note that $T(\alpha u) = \alpha T(u) = \alpha \cdot 0 = 0$. We also can tell that T is injective if and only if $\ker T = \{0\}$. Certainly, if T is injective then $\ker f = \{0\}$ by our previous remarks. Conversely, assume $\ker T = \{0\}$ and let $T(u) = T(v)$. By linearity, we see $0 = T(u) - T(v) = T(u) + T(-v) = T(u + (-v)) = T(u - v)$ whence $u - v \in \ker T$ which implies $u = v$.

A simple example of a linear transformation is the so-called *zero map* which is given by $T(u) = 0$ for all $u \in \mathcal{V}$. Then, $\ker T = \mathcal{V}$ and $\text{Im}(T) = 0$.

PROPOSITION 1.3.12. *Let $\mathcal{B} = \{b_1, b_2, \dots, b_n\}$ be a basis for $\mathcal{V}$ and let $\{t_j\}_{j=1}^n \subseteq \mathcal{W}$. There exists a unique linear map $T : \mathcal{V} \rightarrow \mathcal{W}$ such that $b_j \mapsto t_j$ for all suitable j .*

Proof. Let $T\left(\sum_{j=1}^n \alpha_j b_j\right) = \sum_{j=1}^n \alpha_j t_j$. By uniqueness of representation, it is clear that $b_j \mapsto t_j$ for all suitable j . To see that this is a linear mapping, let $u, v \in \mathcal{V}$. If $\gamma \in \mathbb{F}$, then $\gamma u = \sum_{j=1}^n \gamma \alpha_j b_j$ and hence,

$$T(\gamma u) = \sum_{j=1}^n \gamma \alpha_j t_j = \gamma \sum_{j=1}^n \alpha_j t_j = \gamma T(u)$$

Similarly, by allowing zero coefficients:

$$T(u + v) = T\left(\sum_{j=1}^n (\alpha_j + \beta_j) b_j\right) = \sum_{j=1}^n (\alpha_j + \beta_j) t_j = \sum_{j=1}^n \alpha_j t_j + \sum_{j=1}^n \beta_j t_j = T(u) + T(v)$$

Proving that $T : \mathcal{V} \rightarrow \mathcal{V}$ is a linear transformation. Suppose $L : \mathcal{V} \rightarrow \mathcal{W}$ is another such mapping and let $u \in \mathcal{V}$ be given. u has a unique representation:

$$u = \sum_{j=1}^n \alpha_j b_j$$

Thus,

$$T(u) = \sum_{j=1}^n \alpha_j t_j$$

and by linearity

$$L(u) = L\left(\sum_{j=1}^n \alpha_j b_j\right) = \sum_{j=1}^n L(\alpha_j b_j) = \sum_j \alpha_j L(b_j) = \alpha_j t_j$$

hence, $T \equiv L$ over $\mathcal{V}$.

○

LEMMA 1.3.13. *Let $\mathcal{V}, \mathcal{W}$ be vector spaces over $\mathbb{F}$ and define $\text{Hom}(\mathcal{V}, \mathcal{W}) := \{T : \mathcal{V} \rightarrow \mathcal{W} : T \text{ is linear}\}$. When equipped with standard addition and scalar multiplication this is a vector space. Moreover, if $\mathcal{U}$ is a vector space over $\mathbb{F}$ and we have two mappings:*

$$T : \mathcal{V} \rightarrow \mathcal{W}, \quad L : \mathcal{W} \rightarrow \mathcal{U}$$

both linear transforms, then $L \circ T : \mathcal{V} \rightarrow \mathcal{U}$ is also a linear transformation.

Proof. The fact that this is a vector space is apparent. Let $\alpha \in \mathbb{F}$ be fixed, and let $u, v \in \mathcal{V}$ be given. Then,

$$L \circ T(\alpha u) = L(T(\alpha u)) = L(\alpha T(u)), \quad T(u) \in \mathcal{W}$$

hence, since L is linear we find $L \circ T(\alpha u) = L(\alpha T(u)) = \alpha L(T(u)) = \alpha L \circ T(u)$. Furthermore,

$$L \circ T(u + v) = L(T(u + v)) = L(T(u) + T(v)), \quad T(u), T(v) \in \mathcal{W}$$

whence by linearity of L on $\mathcal{W}$ the result follows.

○

Isomorphisms Let $\mathcal{V}, \mathcal{W}$ be vector spaces over $\mathbb{F}$ and $T : \mathcal{V} \rightarrow \mathcal{W}$ a linear mapping. We shall call this mapping T an *isomorphism* if and only if T is a bijection. We shall say $\mathcal{V}$ is isomorphic to $\mathcal{W}$ if there exists an isomorphism between these spaces, we may then write $\mathcal{V} \cong \mathcal{W}$. The reader should observe being isomorphic is an equivalence relation on vector spaces. In particular, T^{-1} is an isomorphism if and only if T is.

THEOREM 1.3.14. Let $\mathcal{V}$ be a vector space over $\mathbb{F}$ with $\dim(\mathcal{V}) = n < \infty$. Then $\mathcal{V} \cong \mathbb{F}^n$.

Proof. Fix a basis $\mathcal{B} := \{b_j\}_{j=1}^n$ for the space $\mathcal{V}$. Define a mapping:

$$T : \mathcal{V} \rightarrow \mathbb{F}^n, \quad u \mapsto [u]_{\mathcal{B}} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} \quad (1.8)$$

where we write $u = \sum_{j=1}^n \alpha_j b_j$. We claim first that this is a well-defined linear bijection. It is well defined because a vector has a unique representation with respect to the basis. To see that it is linear, we use the property that $[u + v]_{\mathcal{B}} = [u]_{\mathcal{B}} + [v]_{\mathcal{B}}$ and similarly for scalar multiplications.

Now, we see that T is surjective because given $\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}$ one need only take the vector $\sum_{j=1}^n \alpha_j b_j$

in $\mathcal{V}$. Note that $0_{\mathcal{V}} \mapsto 0 \in \mathbb{F}^n$. We now prove that $\ker T = \{0_{\mathcal{V}}\}$, which will establish injectivity.

Certainly, suppose $T(u) = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$. Then $u = \sum_{j=1}^n 0 b_j = 0_{\mathcal{V}}$. This completes the proof.

○

THEOREM 1.3.15. Let $T : \mathcal{V} \rightarrow \mathcal{W}$ be an isomorphism of vector spaces and $\mathcal{B} := \{b_1, b_2, \dots, b_n\}$ a basis for $\mathcal{V}$. The set $\mathcal{C} := \{T(b_1), \dots, T(b_n)\}$ is a basis for $\mathcal{W}$.

Proof. Define $\mathcal{C}$ as given. We shall show first that it is independent. Suppose that $\sum_{j=1}^n \beta_j T(b_j) = 0$. Then, $T\left(\sum_{j=1}^n \beta_j b_j\right) = 0$ which implies that this summation lives in $\ker T$. Since $\mathcal{B}$ is a basis, and hence independent, we deduce that all $\beta_j = 0$ and hence that $\mathcal{C}$ is independent in $\mathcal{W}$. We now show that it is maximally independent.

Else there exists $w \in \mathcal{W}$ so that $\mathcal{C} \cup \{w\}$ is linearly independent. In particular, w cannot be a linear combination of $T(b_j)$ for $j = 1, \dots, n$. But there exists $v \in \mathcal{V}$ so that $T(v) = w$, and we may write

$$v = \sum_{j=1}^n \gamma_j b_j \implies T(v) = w = \sum_{j=1}^n \gamma_j T(b_j)$$

a contradiction.

○

A beautiful corollary is the following:

COROLLARY 1.3.16. Let $\mathcal{V}$ be a vector space of dimension n over $\mathbb{F}$. $\mathbb{F}^n \cong \mathcal{V}$ and two vector spaces $\mathcal{V}, \mathcal{W}$ are isomorphic if and only if $\dim(\mathcal{W}) = \dim(\mathcal{V})$.

THEOREM 1.3.17. Let $T : \mathcal{V} \rightarrow \mathcal{W}$ be a linear mapping with $\dim(\mathcal{V}) < \infty$. Then $\dim(\text{Im}(T)) < \infty$ and moreover,

$$\dim(\mathcal{V}) = \dim(\ker T) + \dim(\text{Im}(T))$$

Proof. Let $\{b_1, \dots, b_n\}$ be a basis for $\ker T$, a subspace of $\mathcal{V}$. By Steinitz' we may extend this to a basis for $\mathcal{V}$: $\{b_1, \dots, b_n, c_1, \dots, c_m\}$. We want to show that $\{c_1, \dots, c_m\}$ is a basis for $\text{Im}(T)$. We shall show that it is minimally spanning. First we show that it is spanning. Let $v \in \text{Im}(T)$, so that there is a linear combination:

$$u = \sum_{j=1}^n \alpha_j b_j + \sum_{j=1}^m \beta_j c_j \mapsto T(u) = v$$

Then,

$$T(u) = \sum_{j=1}^n \alpha_j T(b_j) + \sum_{j=1}^m \beta_j T(c_j) = \sum_{j=1}^m \beta_j T(c_j) = v$$

which confirms that this set is spanning for $Im(T)$. We show it is now minimal. Remove some entry, and after a re-ordering we may suppose it is w_m . Then, $\{w_1, \dots, w_{m-1}\}$ must also span $Im(T)$. In particular, we may write $w_m = \sum_{j=1}^{m-1} \gamma_j w_j$, so that

$$\sum_{j=1}^{m-1} \gamma_j w_j - w_m = 0$$

and hence

○

1.3–1.3.3

References

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- (2) Topology, James Munkres.